Predicates & Quantified Statements II

LECTURE 3.2

CENTRAL QUESTION What exactly would make the claim false? Fix the domain. Identify the quantifier. Test the claim.

INTERPRET & EVALUATE

01 Start with the domain & predicate

Fix a domain D. A predicate P(x) becomes a proposition when x is assigned an element of D.

$\forall x \in D, P(x)$	$\exists x \in D, P(x)$
Every element satisfies P.	At least one witness satisfies P.

Domain contrast: "every x is even".

$D = \{2,4\}$: TRUE; $D = \{1,2,4\}$: FALSE.

$D = \varnothing$: $\forall x \in D\ P(x)$ is T; $\exists x \in D\ P(x)$ is F.

02 Finite domains: AND, OR & De Morgan

For D = {a, b, c}, expand the quantified claim:

$\forall x \in D\ P(x) \equiv P(a) \wedge P(b) \wedge P(c)$

$\exists x \in D\ P(x) \equiv P(a) \vee P(b) \vee P(c)$

D = {1,2,3}; P(x): x is even.

All: $F \wedge T \wedge F = F$ Some: $F \vee T \vee F = T$

$\neg[A \wedge B \wedge C] \equiv \neg A \vee \neg B \vee \neg C$

03 Proof, disproof & representative cases

Worked examples below use the integer domain $\mathbb{Z}$.

Prove ∀	Use an arbitrary n. Every $n^2 \geq 0$. Thus $\forall n \in \mathbb{Z}, n^2 \geq 0$ is true.
Refute ∀	Find one counterexample: n = 0. Thus $\forall n \in \mathbb{Z}, n > 0$ is false.
Prove ∃	Give a witness: n = 2 (also −2). Thus $\exists n \in \mathbb{Z}, n^2 = 4$ is true.
Refute ∃	Rule out every n: n^2 is never negative. Thus $\exists n \in \mathbb{Z}, n^2 = -1$ is false.

A few successful tests do not prove an infinite-domain claim.

NEGATE & CLASSIFY

04 Negate: swap the quantifier

Not all: at least one exception.

$\neg(\forall x \in D\ P(x)) \equiv \exists x \in D\ \neg P(x)$

Not some: no witness exists.

$\neg(\exists x \in D\ P(x)) \equiv \forall x \in D\ \neg P(x)$

Swap ∀/∃; negate the predicate; keep D unchanged.

"Not all passed" means some did not, not that none passed.

05 Negate a universal conditional

$S = \forall x \in D\ (P(x) \to Q(x))$

$\neg S \equiv \exists x \in D\ \neg(P(x) \to Q(x))$

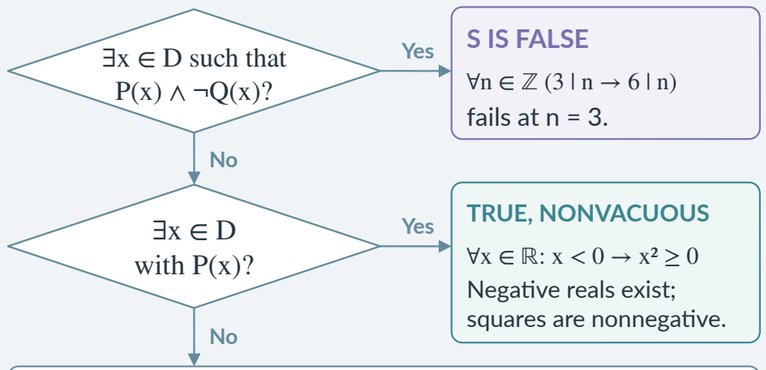
$\equiv \exists x \in D\ (P(x) \wedge \neg Q(x))$

A counterexample satisfies P but fails Q.

If P(a) is false, a does NOT refute S.

06 Classify truth: use the right witness

For $S = \forall x \in D\ (P(x) \to Q(x))$:



"No" must be established over all D, not just tested cases.

COMPARE & APPLY

07 Compare the four conditional forms

Each formula below is universally quantified over D.

ORIGINAL		CONTRAPOSITIVE
$P(x) \to Q(x)$	≡	$\neg Q(x) \to \neg P(x)$
CONVERSE		**INVERSE**
$Q(x) \to P(x)$	≡	$\neg P(x) \to \neg Q(x)$

The two rows are NOT generally equivalent.

Over $\mathbb{Z}$: $4 \mid n \to 2 \mid n$ is true: $n = 4k = 2(2k)$.

Converse: $2 \mid n \to 4 \mid n$ is false at $n = 2$.

08 Necessary, sufficient & only if

For each $x \in D$, abbreviate P(x), Q(x) by P, Q.

P is sufficient for Q	$P \to Q$
P is necessary for Q	$Q \to P$
P only if Q	$P \to Q$
P if Q	$Q \to P$
P is necessary and sufficient for Q	$P \leftrightarrow Q$

$(P \leftrightarrow Q) \equiv (P \to Q) \wedge (Q \to P)$

From P → Q: P suffices for Q; Q is required by P.

The reverse implication needs a separate justification.

09 Apply the logic without overclaiming

Access policy: $Admin(x) \to Access(x)$.

Admin status is sufficient for access; access is necessary for admin status under this policy. Non-admins may have access.

Software: intended $f(n) = n^2$; implementation returns n + n.

At $n = 3$: $6 \neq 9$. A compiling program can be incorrect.

AI: ∀ expresses general rules; ∃ asserts a witness.

Next (3.3): nested quantifiers and why their order matters.

READING KEY Branches group topics. Yes/No arrows classify truth. In formulas: → implication; ≡ logical equivalence.